

***Generation Interconnection
System Impact Study Report***

For

***PJM Generation Interconnection Request
Queue Position AD1-069***

Iron Bridge – Donegal 138 kV

October 2018

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

General

EverPower Wind Development, LLC (“Interconnection Customer”) has proposed a new wind generation facility in the Fayette County, Pennsylvania. The installed facilities will have capability of 80 MW with 11.76 MW of this output being recognized by PJM for Capacity Interconnection Rights. The proposed in-service date is December 15, 2020. This study does not imply a **West Penn Power (“Transmission Owner”) commitment to this in-service date.**

Point of Interconnection (“POI”)

This project will interconnect with the West Penn Power transmission system by tapping the Iron Bridge – Donegal 138 kV line at a point located approximately 5.2 miles from Iron Bridge substation. This transmission tap will be facilitated by building a new three breaker ring bus switching station and a one span loop of the Iron Bridge – Donegal transmission line. Please refer to Appendix 2 for one-line diagram of system configuration.

Schedule

Based on the extent of the Transmission Owner attachment facilities and network upgrades required to support the AD1-069 generation project, it is expected to take a minimum of twenty three (23) months from the date of a fully executed Interconnection Construction Service Agreement to complete the installation. This includes the requirement for Interconnection Customer to make a preliminary payment to FirstEnergy (via PJM) which funds the first three months of engineering design that is related to the construction of the Direct Connection facilities. It is assumed that Interconnection Customer will provide the property for the AD1-069 138 kV interconnection substation and all rights-of-way, permits, easements, etc. that will be needed. A further assumption is that there will be no environmental issues with any of the new properties associated with this project, that there will be no delays in acquiring the necessary permits for implementing the defined network upgrades, and that all system outages will be allowed when requested.

Transmission Owner Scope of Work and Costs Summary

The total estimated cost of the direct and non-direct connection facilities is \$8,158,800. This cost does not include a Contribution in Aid of Construction (“CIAC”) Federal Income Tax Gross Up charge of \$1,452,500. This tax may or may not be charged based on whether this project meets the eligibility requirements of IRS Notice 88-129.

Please Note: Albeit the fact that there is no system reinforcement costs listed below for the project (AD1-069), the summer peak load flow study identified constraints which will be relieved by baseline upgrades. The existing PJM baseline project to reconductor line (b2965). This project has an in-service date of 06/01/2022. The estimated cost \$7.08 Millions. The interconnection of AD1-069 is depending on this baseline upgrade project and an interim study could be required. Interconnection Customer will be responsible for the cost of accelerating the project to 2020. Please refer to the network impacts section of this report for more details.

All PJM network upgrade numbers will be determined at the facilities study stage.

The following upgrades are required to support AD1-069 Interconnection.

a) Attachment Facilities: None.

b) Direct Connection Network Upgrades

- b1) Construct a three breaker 138kV ring bus interconnection substation with terminals to Donegal, Iron Bridge and AD1-069 generation.....**\$5,784,700.**
- b2) Loop the Iron Bridge-Donegal 138kV line to the new AD1-069 substation.**\$714,900**
- b3) Modify nameplates and drawings at Donegal substation:**\$18,500**
- b4) Modify nameplates and drawings at new switching substation:.....**\$18,500**
- b5) Project Management, Commissioning, Environmental, Forestry, Real Estate, Right of Way, and SCADA.:.....**\$626,800**

c) Non-Direct Network Upgrades:

- c1) At Iron Bridge Substation: Replace 138kV AD1-069 (Donegal) line breaker, leads, and all relaying and control cables. Add 138kV line trap, CVT and arresters.**\$837,800**
- c2) Estimated Multi-Protocol Label Switching (“MPLS”) router at AD1-069 Interconnection substation to provide SCADA transport for new Remote Terminal Unit (“RTU”). Next MPLS hop for SCADA backhaul is assumed to be at Pleasant Valley substation. Estimated cost:**\$157,800**

d) Direct Local Network Upgrades: None.

e) Non-Direct Local Network Upgrades: None.

f) Option to Build Upgrades: None.

Estimated Total Costs (a) to (f):\$8,158,800

Interconnection Customer Requirements

The proposed Customer Facilities must be designed in accordance with FirstEnergy's "Requirements for Transmission Connected Facilities" document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>. In particular, the Interconnection Customer is responsible for the following:

1. The purchase and installation of fully rated 138 kV circuit breaker to protect the AD1-069 generator lead line. A single circuit breaker must be used to protect this line; if the project has several GSU transformers, the individual GSU transformer breakers cannot be used to protect this line.
2. The purchase and installation of the minimum required FirstEnergy generation interconnection relaying and control facilities. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.
3. The purchase and installation of supervisory control and data acquisition (SCADA) equipment to provide information in a compatible format to the FirstEnergy Transmission System Control Center.
4. A compliance with the FirstEnergy and PJM generator power factor and voltage control requirements.
5. The execution of a back-up service agreement to serve the customer load supplied from the AD1-069 generation project metering point when the units are out-of-service. This assumes the intent of Interconnection Customer is to net the generation with the load.
6. Power Factor Requirements: Interconnection Customer shall design its non-synchronous Customer Facility with the ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.95 lagging (supplying VARs) measured at the high-side of the facility substation transformer(s) connected to the FirstEnergy transmission system.
7. System Protection Requirements: Interconnection Customer must design its Customer Facilities in accordance with all applicable standards, including the standards in FirstEnergy's "Requirements for Transmission Connected Facilities" document located at above shown link. Preliminary Protection requirements will be provided as part of the Facilities Study. Detailed Protection Requirements will be provided once the project enters the construction phase.
8. Revenue Metering and SCADA Requirements: PJM requires that Interconnection Customer must install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for Interconnection Customer's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Sections 24.1 and 24.2. FirstEnergy requires that Interconnection Customer must comply with all FirstEnergy Revenue Metering Requirements for Generation Interconnection Customers. The Revenue Metering Requirements may be found within the "FirstEnergy Requirements for Transmission Connected Facilities" document located at the following links: <http://www.firstenergycorp.com/feconnect> and <http://www.pjm.com/planning/design-engineering/to-tech-standards.aspx>
9. Interconnection Customer must meet all PJM, ReliabilityFirst and NERC reliability criteria and operating procedures required for standards compliance. For example, Interconnection Customer will need to properly locate and report the over and under-voltage and over and under-frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and ReliabilityFirst audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the FirstEnergy system.

The above requirements apply in addition to any metering or other requirements set by PJM.

Network Impacts

The Queue Project AD1-069 was evaluated as a 80.0 MW (Capacity 11.8 MW) injection tapping Iron Bridge to Donegal 138kV line in the APS area. Project AD1-069 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AD1-069 was studied with a commercial probability of 53%. Potential network impacts were as follows:

Summer Peak Analysis - 2021

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

1. (AP - AP) The 01ALLENP-01CHARLR 138 kV line (from bus 235135 to bus 235161 ckt 1) loads from 112.09% to 119.78% (**DC power flow**) of its emergency rating (179 MVA) for the tower line contingency outage of 'AP-P7-1-WPP-138-80'. This project contributes approximately 13.77 MW to the thermal violation.

CONTINGENCY 'AP-P7-1-WPP-138-80' /* 58
DISCONNECT BRANCH FROM BUS 235161 TO BUS 235277 CKT 1 /* 01CHARLR 138 01YUKON 138
DISCONNECT BRANCH FROM BUS 235270 TO BUS 235277 CKT 1 /* 01WESTRV 138 01YUKON 138
END

Please refer to Appendix 3 for a table containing the generators having contribution to this flowgate.

2. (AP - AP) The 01ALLENP-01CHARLR 138 kV line (from bus 235135 to bus 235161 ckt 1) loads from 111.14% to 118.83% (**DC power flow**) of its emergency rating (179 MVA) for the tower line contingency outage of 'AP-P7-1-WPP-138-78'. This project contributes approximately 13.77 MW to the thermal violation.

CONTINGENCY 'AP-P7-1-WPP-138-78' /* 57
DISCONNECT BRANCH FROM BUS 235161 TO BUS 235270 CKT 1 /* 01CHARLR 138 01WESTRV 138
DISCONNECT BRANCH FROM BUS 235161 TO BUS 235277 CKT 1 /* 01CHARLR 138 01YUKON 138
END

3. (AP - AP) The 01YUKON-01SMTH62 138 kV line (from bus 235277 to bus 235252 ckt 1) loads from 101.62% to 102.35% (**DC power flow**) of its emergency rating (376 MVA) for the line fault with failed breaker contingency outage of 'AP-P2-3-WP-138-204'. This project contributes approximately 6.1 MW to the thermal violation.

CONTINGENCY 'AP-P2-3-WP-138-204'	/* 214	
DISCONNECT BRANCH FROM BUS 235135 TO BUS 235161 CKT 1		/* 01ALLENP 138 01CHARLR 138
DISCONNECT BRANCH FROM BUS 235155 TO BUS 235161 CKT 1		/* 01CALIFN 138 01CHARLR 138
DISCONNECT BRANCH FROM BUS 235161 TO BUS 235270 CKT 1		/* 01CHARLR 138 01WESTRV 138
DISCONNECT BRANCH FROM BUS 235161 TO BUS 235260 CKT 1		/* 01CHARLR 138 01UNIONJ 138
END		

Please refer to Appendix 4 for a table containing the generators having contribution to this flowgate.

4. (AP - AP) The 01YUKON-01SMTH62 138 kV line (from bus 235277 to bus 235252 ckt 1) loads from 101.45% to 102.19% (**DC power flow**) of its emergency rating (376 MVA) for the bus fault outage of 'AP-P2-2-WP-138-019'. This project contributes approximately 6.1 MW to the thermal violation.

CONTINGENCY 'AP-P2-2-WP-138-019'	/* CHARLEROI #1 138KV BUS
DISCONNECT BRANCH FROM BUS 235161 TO BUS 235260 CKT 1	/* 01CHARLR 138 01UNIONJ 138
DISCONNECT BRANCH FROM BUS 235161 TO BUS 235155 CKT 1	/* 01CHARLR 138 01CALIFN 138
DISCONNECT BRANCH FROM BUS 235161 TO BUS 235135 CKT 1	/* 01CHARLR 138 01ALLENP 138
DISCONNECT BRANCH FROM BUS 235161 TO BUS 235270 CKT 1	/* 01CHARLR 138 01WESTRV 138
DISCONNECT BRANCH FROM BUS 235161 TO BUS 235872 CKT 1	/* 01CHARLR 138 01CHAR 25
END	

Steady-State Voltage Requirements

To be determined during later study phases.

Short Circuit

None

Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Only the most severely overloaded conditions are listed. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed, which will study all overload conditions associated with the overloaded element(s) identified.

None

Light Load Analysis - 2021

To be determined during the facilities study stage.

System Reinforcements

Short Circuit

None

Stability and Reactive Power Requirement

To be determined during the facilities study stage.

Summer Peak Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

1. (AP - AP) The 01ALLENP-01CHARLR 138 kV line (from bus 235135 to bus 235161 ckt 1) loads from 112.09% to 119.78% (**DC power flow**) of its emergency rating (179 MVA) for the tower line contingency outage of 'AP-P7-1-WPP-138-80'. This project contributes approximately 13.77 MW to the thermal violation.

APS:

Existing PJM baseline project to reconductor line (b2965). This project has an in-service date of 06/01/2022. The estimated cost \$7.08 M. Customer responsible for the cost of accelerating the project to 2021.

2. (AP - AP) The 01ALLENP-01CHARLR 138 kV line (from bus 235135 to bus 235161 ckt 1) loads from 111.14% to 118.83% (**DC power flow**) of its emergency rating (179 MVA) for the tower line contingency outage of 'AP-P7-1-WPP-138-78'. This project contributes approximately 13.77 MW to the thermal violation.

Same as Contribution to Previously Identified #1

3. (AP - AP) The 01YUKON-01SMTH62 138 kV line (from bus 235277 to bus 235252 ckt 1) loads from 101.62% to 102.35% (**DC power flow**) of its emergency rating (376 MVA) for the line fault with failed breaker contingency outage of 'AP-P2-3-WP-138-204'. This project contributes approximately 6.1 MW to the thermal violation.

APS:

Existing PJM baseline project to reconductor line (b2966). This project has an in-service date of 06/01/2022. The estimated cost \$3.19 M. Customer responsible for the cost of accelerating the project to 2021.

4. (AP - AP) The 01YUKON-01SMTH62 138 kV line (from bus 235277 to bus 235252 ckt 1) loads from 101.45% to 102.19% (**DC power flow**) of its emergency rating (376 MVA) for the bus fault outage of 'AP-P2-2-WP-138-019'. This project contributes approximately 6.1 MW to the thermal violation.

Same as Contribution to Previously Identified #3

Light Load Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

To be determined during the facilities study stage.

Winter Peak Analysis - 2021

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

Steady-State Voltage Requirements

None

Winter Peak Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

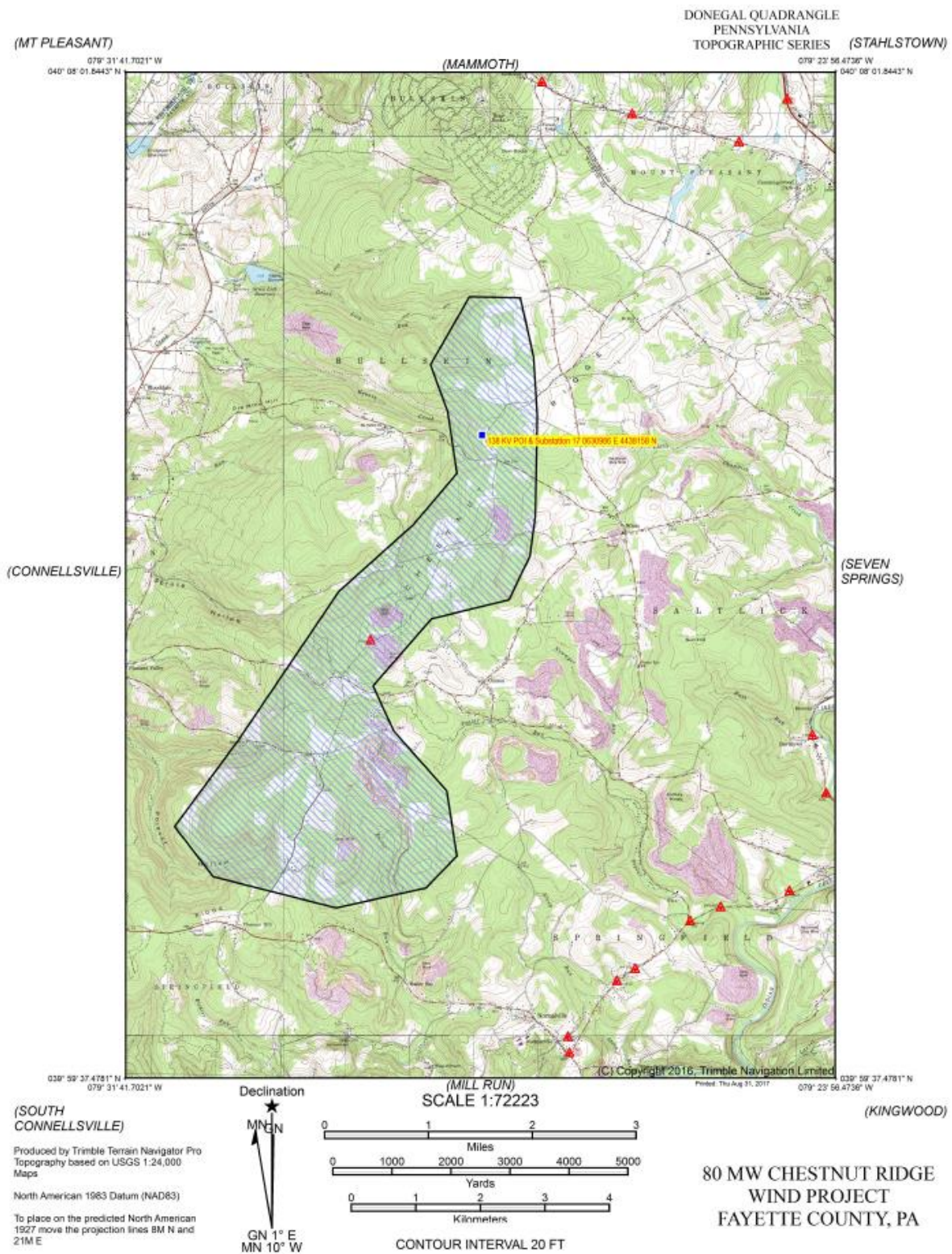
(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

None

Appendix 1

Facility Location

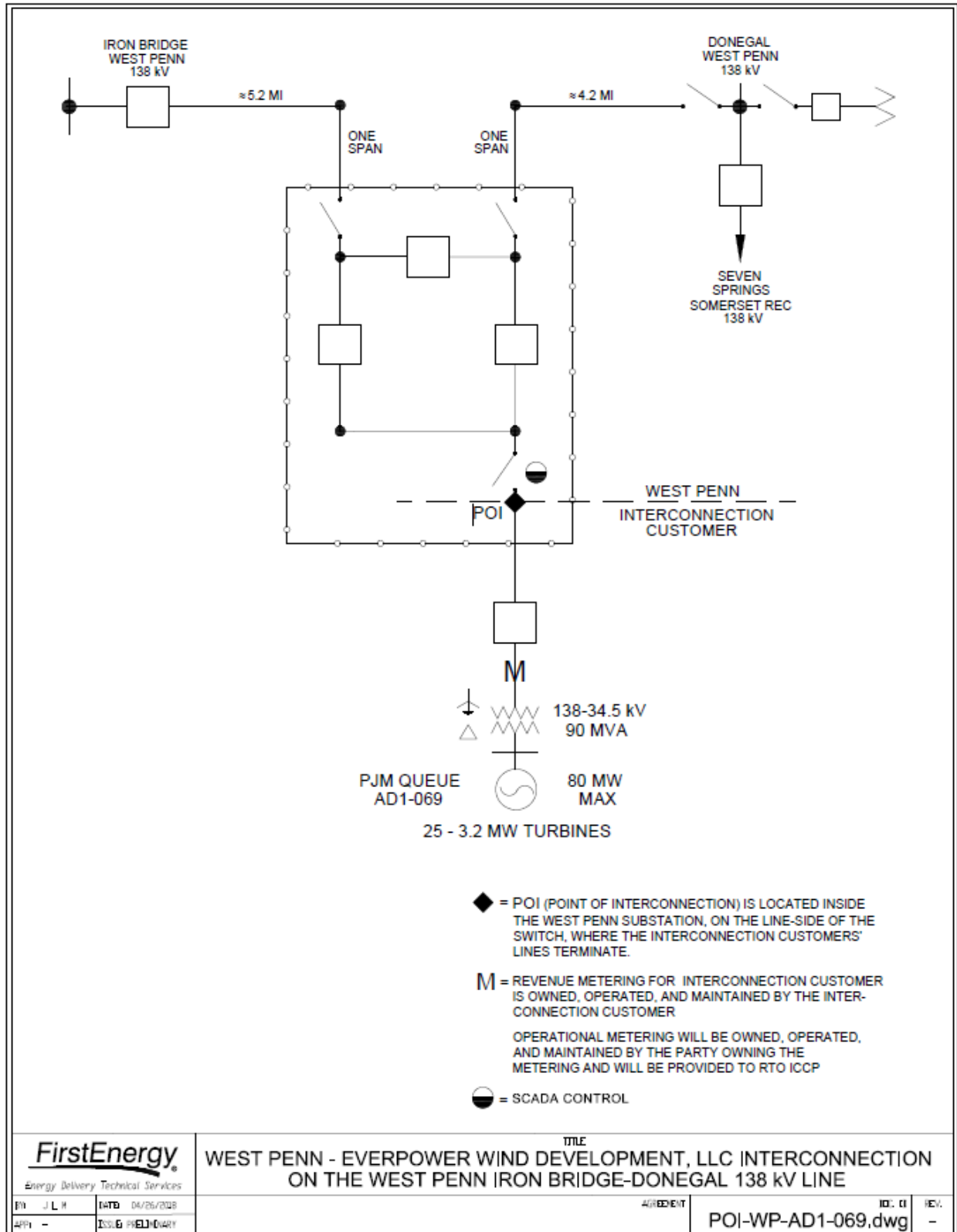
PJM Queue Position: AD1-069



Appendix 2

Interconnection One-Line Diagram

PJM Queue Position: AD1-069



Appendix 3

Flowgate Information: Allenport – Charleroi 138 kV Line PJM Queue Position: AD1-069

This appendix contains additional information about the flowgate presented in the body of the report. The intent of this appendix section is to provide more information on which projects/generators have contributions to the flowgate in question. Although this information is not used "as is" for cost allocation purposes, it can be used to gauge other generators impact. It should be noted the generator contributions presented in the appendices sections are full contributions, whereas in the body of the report, those contributions take into consideration the commercial probability of each project.

(AP - AP) The 01ALLENP-01CHARLR 138 kV line (from bus 235135 to bus 235161 ckt 1) loads from 112.09% to 119.78% (**DC power flow**) of its emergency rating (179 MVA) for the tower line contingency outage of 'AP-P7-1-WPP-138-80'. This project contributes approximately 13.77 MW to the thermal violation.

CONTINGENCY 'AP-P7-1-WPP-138-80'
/* 58

DISCONNECT BRANCH FROM BUS
235161 TO BUS 235277 CKT 1 /*
01CHARLR 138 01YUKON 138

DISCONNECT BRANCH FROM BUS
235270 TO BUS 235277 CKT 1 /*
01WESTRV 138 01YUKON 138

END

Bus Number	Bus Name	Full Contribution
236523	01WDPWR_K20	0.04
920571	AA2-161 C	26.98
920572	AA2-161 E	1.47
934451	AD1-069 C	2.02
934452	AD1-069 E	11.74

Appendix 4

Flowgate Information: Yukon – Smithton 138 kV 62 Line PJM Queue Position: AD1-069

This appendix contains additional information about the flowgate presented in the body of the report. The intent of this appendix section is to provide more information on which projects/generators have contributions to the flowgate in question. Although this information is not used "as is" for cost allocation purposes, it can be used to gauge other generators impact. It should be noted the generator contributions presented in the appendices sections are full contributions, whereas in the body of the report, those contributions take into consideration the commercial probability of each project.

(AP - AP) The 01YUKON-01SMTH62 138 kV line (from bus 235277 to bus 235252 ckt 1) loads from 101.62% to 102.35% (**DC power flow**) of its emergency rating (376 MVA) for the line fault with failed breaker contingency outage of 'AP-P2-3-WP-138-204'. This project contributes approximately 6.1 MW to the thermal violation.

CONTINGENCY 'AP-P2-3-WP-138-204'
/* 214

DISCONNECT BRANCH FROM BUS
235135 TO BUS 235161 CKT 1 /*
01ALLENP 138 01CHARLR 138

DISCONNECT BRANCH FROM BUS
235155 TO BUS 235161 CKT 1 /*
01CALIFN 138 01CHARLR 138

DISCONNECT BRANCH FROM BUS
235161 TO BUS 235270 CKT 1 /*
01CHARLR 138 01WESTRV 138

DISCONNECT BRANCH FROM BUS
235161 TO BUS 235260 CKT 1 /*
01CHARLR 138 01UNIONJ 138

END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
915692	Y3-103 E	-5.48
920571	AA2-161 C	47.36
920572	AA2-161 E	2.58
934451	AD1-069 C	0.9
934452	AD1-069 E	5.2